UK Data Service Discover: Visible Connections and a Structuralist Approach to Discovery/Making Data Visible - Building an Enterprise Search Solution from the Ground Up

Lucy Bell, Matthew Brumpton

UK Data Archive

IASSIST 2013, Cologne

29 May 2013

UK Data Service

Discover: visible data

- The vision, background and context (Lucy)
- The metadata challenges and solutions (Lucy)
- The technology (Matthew)

Discover: the vision

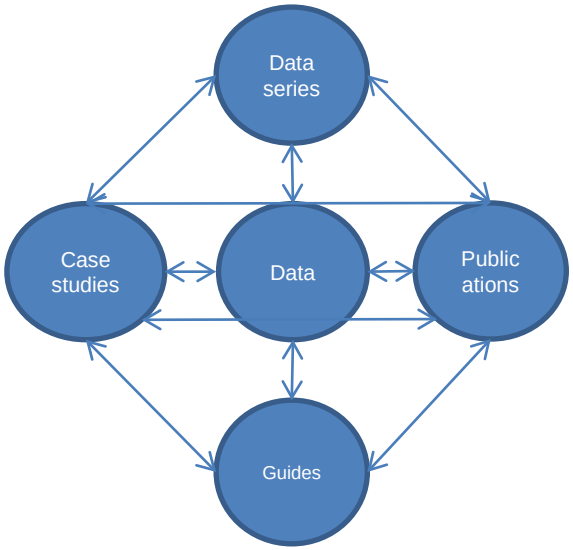
- 2010 – 2013: review, make recommendations for and then develop a new approach to resource discovery for the UK Data Service
- the result:
  - **Discover** – a faceted search and browse application, which capitalises on the rich, DDI metadata, and controlled vocabularies, used to index data collections, case studies, support guides and publications

Discover: the background

- The review asked how researchers find data:
  - trends in information-seeking behaviour showed that users prefer simple, Google-like interfaces…
  - …but which still return acutely-focused and highly-relevant results
  - the look and feel of the interfaces should be simple but the results must achieve academic rigour
- so metadata are required which are both **extremely powerful** AND **increasingly invisible**

Discover: the wider context

- Functional Requirements for Bibliographic Records (FRBR) takes a holistic, structuralist view of metadata
- all items are seen in the context of the whole, with their connections and relationships used to navigate the world they, as a group, describe
- we have applied these principles to Discover, through which users may ‘discover’:

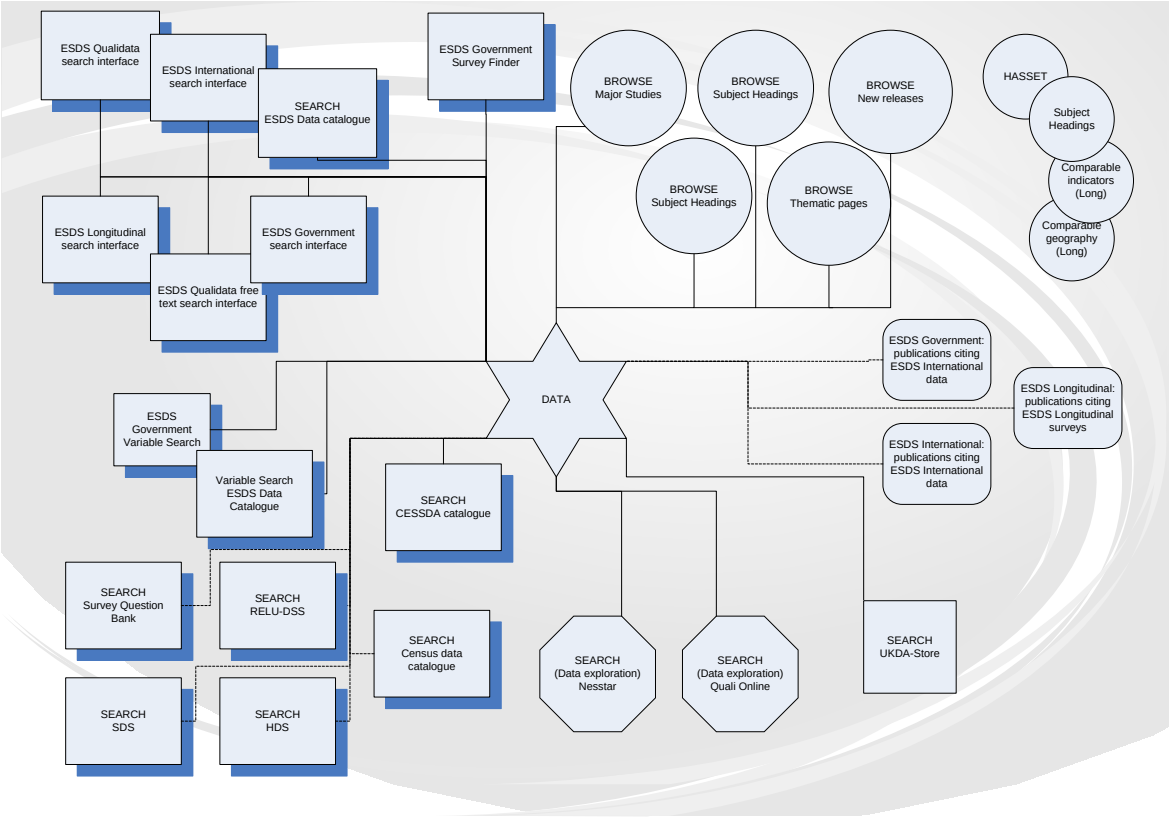


The resource discovery can take place through a simple search, a faceted browse, or by navigating the resources’ relationships. More navigations are planned.

UK Data Service

Discover: the local context

- Previously we had a federated search environment



- with 21 interfaces/applications

UK Data Service

Discover: the solution

- a single search interface which allows the user to move seamlessly from one type of resource to another
  - via faceted browsing
  - and directly from within each resource type

- this required **cross-referencing** data collections with:
  - publications
  - research outputs
  - support guides
  - case studies
  - (and more cross-references are planned)
- AND the use of many **controlled vocabularies**

Discover: the metadata challenges

- facets needing controlled vocabularies
- some were already in a fit state
  - Country <nation>
  - Depositor <depositr> (forthcoming)
- others needed mapping to high levels:
  - Subject categories <topcClas>
  - Access <specPerm>
- many were populated with free text:
  - Analysis unit <anlyUnit>
  - Spatial unit <geogUnit>
  - Kind of data <dataKind>

Discover: the metadata challenges (2)

- the principles:
  - manipulate and compare metadata per element, using Google Refine and Excel
  - map to existing CV where possible (e.g. spatial units list created under a separate project, U.Geo; or DDI <dataKind> CVG's list)
  - in some cases, create a CV
  - re-import 'clean' metadata to the indexes
  - CVs to be used in the future

- Over summer 2012 it took 2 staff members, working c.0.4 FTE 4 months to clean 3 elements

Discover: the challenges (3)

- the mappings:

Facet	DDI element	Existing unique values	Mapped to a CV of
Access	<specPerm>	30	9
Subject	<topcClas>	116	21 (TTs)
Spatial unit	<geogUnit>	653	194
Analysis unit	<anlyUnit>	183	11
Kind of data	<dataKind>	294	7

- plus a new CV:
  - Data type (longitudinal, qualitative/mixed, microdata etc.)

- ideas of more to come….
  - Method of data collection <collMode>
  - Method of access (Explore online or download)

Discover: the technology

Data

- MS SQL
- XML (Derived from DDI 2.1)
- ESRC OAI Feeds

Search Engine

- Solr 4.1
- Tomcat 7
- Java 7
- Windows 2008 R2

Web Application

- HTML, JavaScript, CSS
- .NET 4.0
- MS SQL

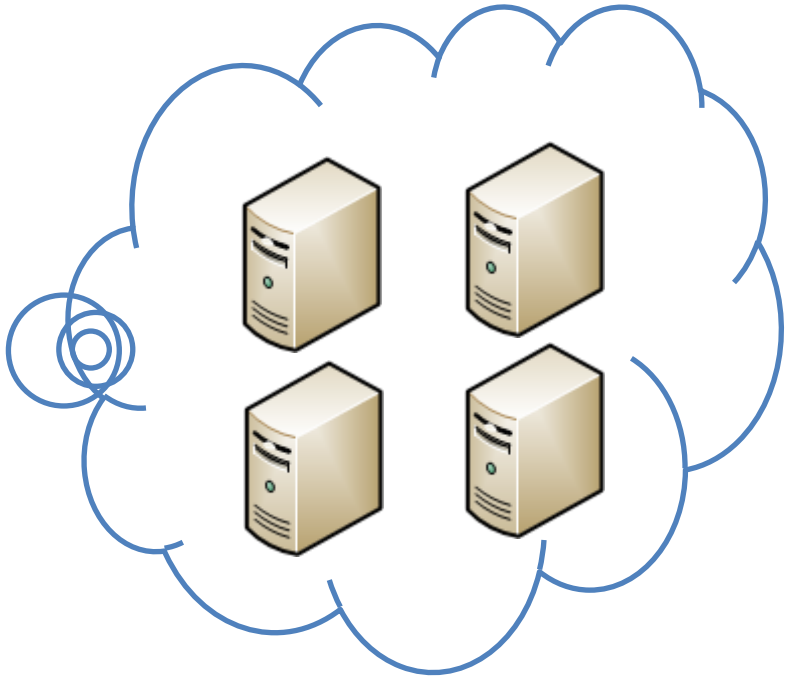
Discover: the data

- Data types
  - Date, string, integer
  - Facet, multi-value
- Shared meta data
  - Mapping
- Cleansing
  - Dates
  - HTML / XML

Data collections	ESRC Outputs	User guides	Case Studies
Type Subject Date	Type Subject Date	Type Subject Date	Type Subject Date
Data type Key data Country Kind of data Spatial unit Analysis Access	Source Author Type of output	Type of guide	Course level Course type Data type Case study type

Discover: Solr

- Open source solution written in Java
- Configured by XML
- Features include:
  - Full-text search
  - Faceted search
  - Highlighting
  - Spell check
  - Rich document handling
- Daily index
- Windows servers
- Challenges
  - Open source
  - Up skilling

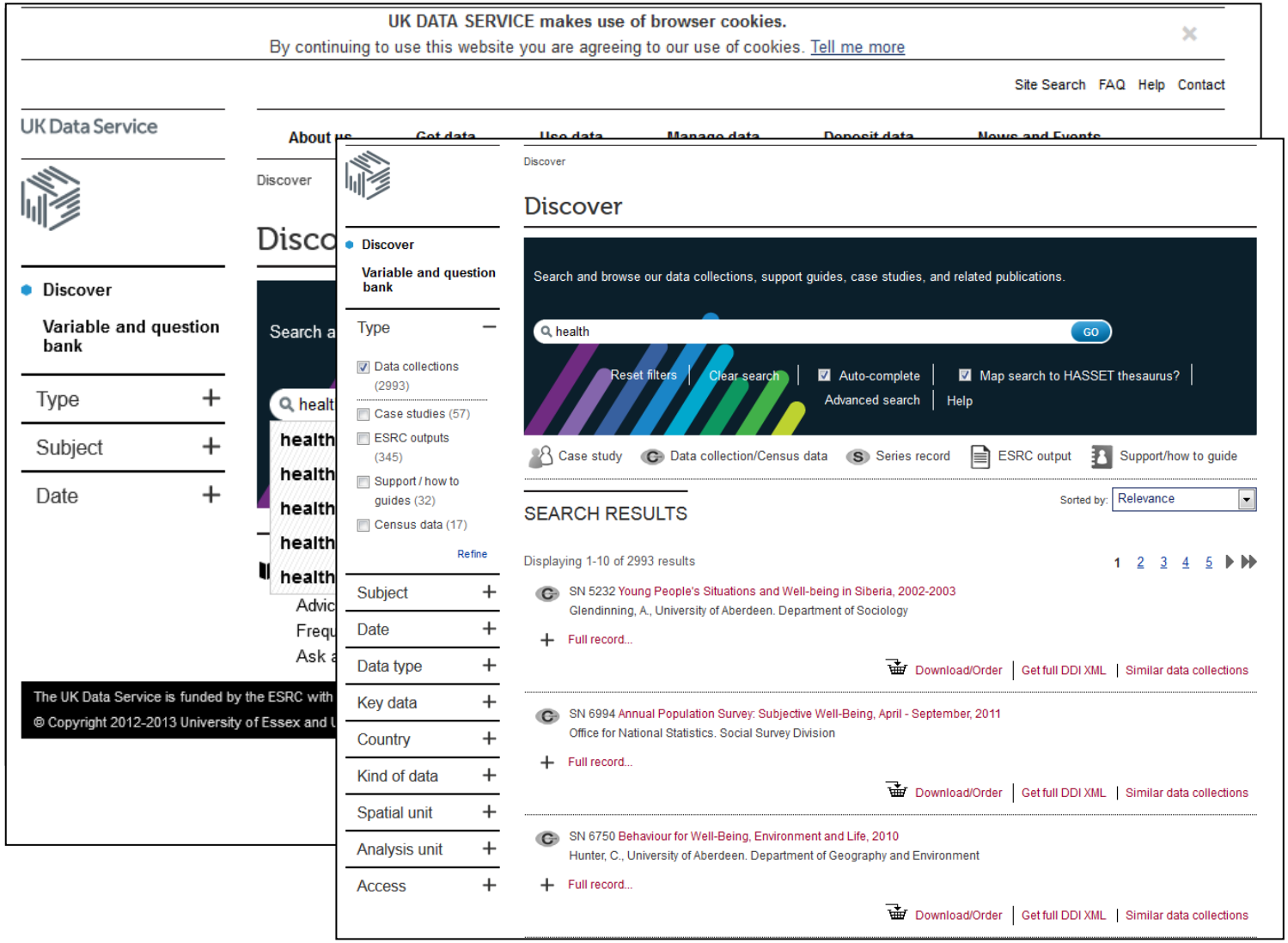


Cloud configuration (4 production servers)

Discover: the application

- Supports a wide range of browsers and devices
  - Twitter Bootstrap - Grid system and responsive design
  - Modernizer - Feature detection
- Warning!
  - Responsive design only hides content
  - Dedicated mobile pages required to reduce load
- Page load times under 1 second
  - Minimise the data being returned
  - Use of asynchronous requests

Discover: the result

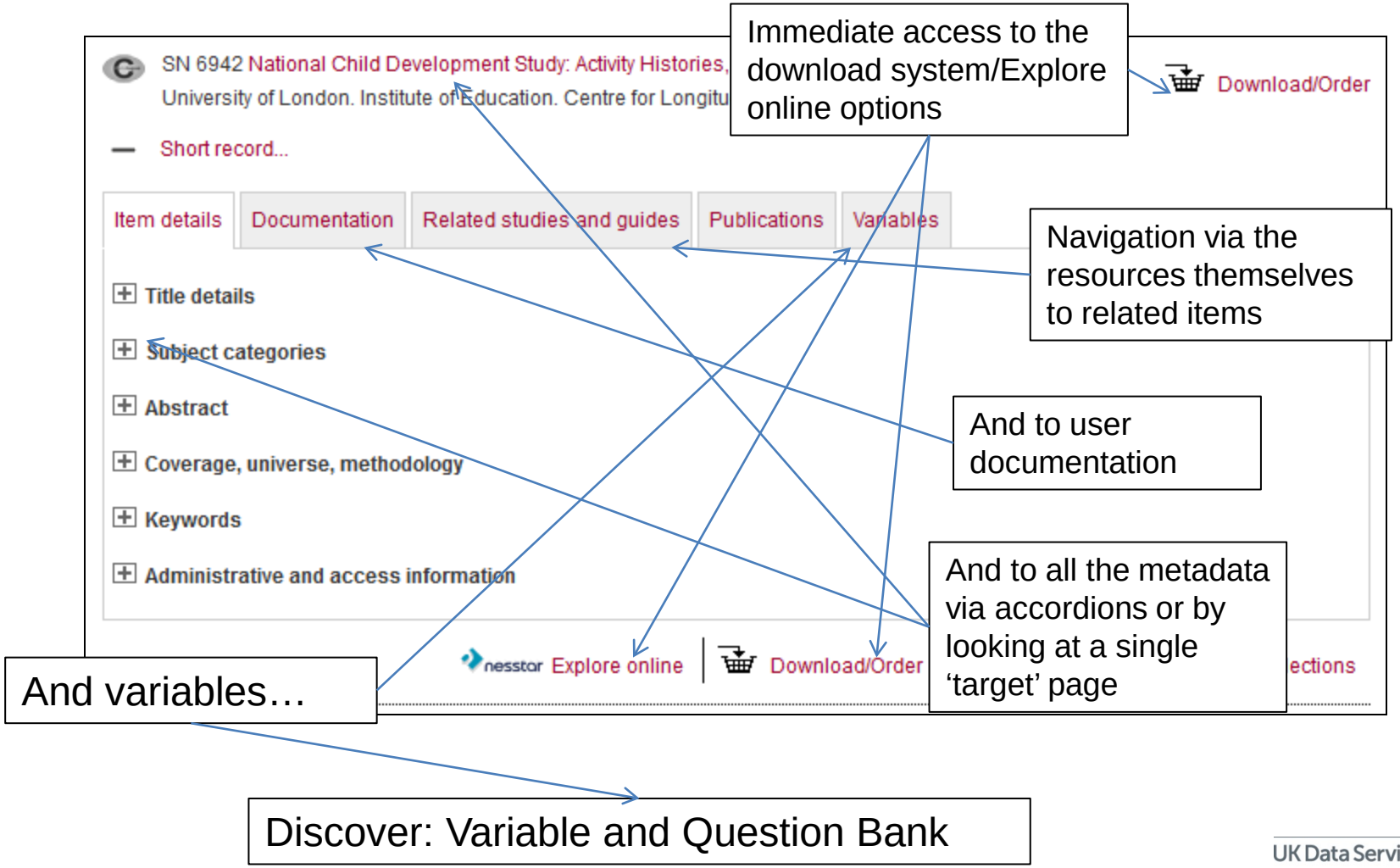


Simple searches, plus suggestions and help

Or browsing using facets

UK Data Service

Discover: the result (2)



UK Data Service

Discover: the result (2)

SN 6942 National Child Development Study: Acti
University of London. Institute of Education. Cent

Download/Order

Short record...

Item details | Documentation | Related studies and

Related studies

Related case studies

Do comprehensive schools reduce social mobility?
Does being left-handed or right-handed affect academ
Adult education and its effect on heart disease
Does childhood poverty affect respiratory health in mic

Related support guides

nesstar Explo

data collections

Case study - research

Does being left-handed or right-handed affect academic ability?

Author: Christopher Cheyne, University of Liverpool, Marta Garcia-Finana, University of Liverpool, Neil Roberts, University of Edinburgh, Tim J. Crow, University of Oxford, and Stuart J. Leask, University of Nottingham.

Date: 5 November 2011

Type of case study: Research

About the research

You are a left-handed girl. Does that predict better grades than a right-handed boy?

Stories and superstitions about 'lefties' have been around for generations. In recent years, there has been a special curiosity about whether there are links between handedness and academic abilities. In this study the authors investigated the effects of gender, writing hand, relative hand skill, and UK region on mathematics and reading test scores from pupils across Britain.

To do this, the researchers analysed responses from 11-year olds included in the National Child Development Study to find associations between cognitive ability (reading and maths scores) and a range of factors including handedness and gender.

Their findings indicate that children who write with one hand, while being better skilled with the other hand, performed worse in both the reading and maths tests than children who write with the hand which has the most hand skill, irrespective of whether the child was left- or right-handed.

This suggests that handedness tasks should be considered in conjunction with a measure of hand skill in future studies of investigating cognitive ability and handedness.

About the data

The research relied on data from the National Child Development Study (NCDS), a continuing longitudinal study that follows the lives of all those living in Great Britain who

UK Data Service

Discover: the result (3)

Site Search FAQ Help Contact

UK Data Service

About us | Get data | Use data | Manage data | Deposit data | News and Events

Discover > Variables

Variable and question bank

My Variables

Discover

- Variable and question bank

Survey +

Country +

Date +

Data collector +

Harmonised question? +

Question type +

Search and browse variables and questions from survey datasets.

Search our variables GO

Reset filters | Clear search | Search only question text/responses | Auto-complete | Help

NEW USERS

Advice for new users
Frequently asked questions
Ask a question

MOST POPULAR PAGES

Discover data and resources
Data by type
How to access data

CONNECT WITH US

Email Twitter Facebook

The UK Data Service is funded by the ESRC with contributions from the University of Essex and the University of Manchester.

Home Terms and conditions Cookies

© Copyright 2012-2013 University of Essex and University of Manchester. All rights reserved.

UK Data Service

Questions

- http://discover.ukdataservice.ac.uk